1.0 INTRODUCTION

The Titanic dataset contains information about passengers aboard the RMS Titanic, including their survival status, demographic details, ticket class, fare price, and port of embarkation.

The purpose of this analysis is to explore the dataset and extract meaningful insights about the factors that influenced passenger survival. Using data manipulation techniques learned in Lab 7a, the dataset was imported, cleaned, filtered, and analysed to identify patterns and relationships within the data.

2.0 DATASET DESCRIPTION

The dataset used in this analysis is titanic.csv, which contains information about passengers aboard the Titanic. This dataset helps analyze survival patterns based on demographic and socio-economic factors.

Key Variables:

Variable	**Description**
Survived	Survival status (0 = No, 1 = Yes)
Pclass	Passenger class (1st, 2nd, 3rd)
Sex	Passenger gender
Age	Passenger age
SibSp	Number of siblings/spouses aboard
Parch	Number of parents/children aboard
Fare	Ticket fare price
Embarked	Port where passenger boarded

3.0 DATA PREPARATION

Before analysis, the dataset was processed using several data management techniques.

3.1 Importing Dataset

The dataset was imported into R using the read.csv() function.

3.2 Identifying Missing Values

Missing values were identified using:

- is.na()
- sum(is.na())

- sapply() to count NA values in each column.

3.3 Handling Missing Values

Rows containing missing values were removed using na.omit(). This ensured that the dataset used for analysis contained only complete records.

4.0 DATA EXPLORATION AND FILTERING

The **dplyr library** was used to filter and analyze the dataset.

Filtering techniques included:

- Selecting passengers based on gender
- Selecting passengers with high ticket fare
- Filtering using multiple conditions
- Sorting passengers based on ticket price

Examples of filtering operations include:

- Female passengers
- Passengers with fare greater than 500
- Female passengers with high ticket fares

Sorting techniques were also applied using the arrange() function to organize passengers by fare values.

5.0 DATA ANALYSIS AND INSIGHTS

Several observations were derived from the dataset after performing filtering and sorting operations.

5.1 General Observations of Dataset

Before performing detailed analysis, several general observations can be made about the Titanic dataset. Understanding the structure and meaning of the variables is important in identifying which factors may influence passenger survival.

- The most important column in the dataset is the **Survived** variable. This column represents whether a passenger survived the Titanic disaster, where the value **1 indicates survival and 0 indicates the passenger did not survive**. This variable serves as the primary outcome variable for the analysis because it allows us to evaluate which passenger characteristics may have influenced survival.

- Another important variable is **Sex**, which indicates the gender of each passenger. This variable allows us to examine whether survival rates differed between male and female passengers. Since historical records suggest that evacuation policies prioritised women and children, analysing survival based on gender can provide meaningful insights into how rescue efforts were conducted.

- The **Pclass (Passenger Class)** variable represents the socio-economic class of passengers, categorized into **First Class, Second Class, and Third Class**. Passenger class is particularly important because it may influence a passenger's location on the ship, access to lifeboats, and overall chances of survival. Higher class passengers may have had better access to safety measures compared to passengers in lower classes.

- The **Fare** variable provides information about the ticket price paid by passengers. Since ticket prices are often related to passenger class and cabin location, this variable can be used to examine whether passengers who paid higher fares had higher survival probabilities.

- The **Embarked** variable indicates the port where passengers boarded the Titanic, such as **Cherbourg, Queenstown, and Southampton**. This variable is useful in understanding whether passengers from certain embarkation locations had different survival outcomes, which may reflect differences in passenger demographics or class distribution.

- In addition, the dataset contains **family-related variables**, including **SibSp** (number of siblings or spouses aboard) and **Parch** (number of parents or children aboard). These variables can help identify whether travelling alone or with family members influenced a passenger's ability to survive during the evacuation.

Overall, these variables provide a combination of **demographic, economic, and social factors** that can be analysed to better understand patterns of survival during the Titanic disaster.

5.2 Survival Rate by Gender

```
> titanic_cleaned %>%
+     group_by(Sex) %>%
+     summarise(
+         Total = n(),
+         Survived = sum(Survived),
+         Survival_Rate = round(mean(Survived)*100,2)
+     )
# A tibble: 2 × 4
  Sex    Total Survived Survival_Rate
  <chr>  <int>    <int>         <dbl>
1 female   261      197         19700
2 male     453       93          9300
```

The analysis shows that female passengers had a significantly higher survival rate than male passengers.

Insight:

Women were more likely to survive due to the evacuation policy commonly described as "women and children first."

5.3 Survival Rate by Passenger Class

```
> titanic_cleaned %>%
+     group_by(Pclass) %>%
+     summarise(
+         Total = n(),
+         Survived = sum(Survived),
+         Survival_Rate = round(mean(Survived)*100,2)
+     )
# A tibble: 3 × 4
  Pclass Total Survived Survival_Rate
   <int> <int>    <int>         <dbl>
1      1   186      122         12200
2      2   173       83          8300
3      3   355       85          8500
```

Passengers travelling in First Class had the highest survival rate, while Third Class passengers had the lowest.

Insight:

Passengers from higher social classes had better access to lifeboats, increasing their survival probability.

5.4 Survival Based on Fare Price

```
> titanic_cleaned %>%
+     group_by(Survived) %>%
+     summarise(
+         Average_Fare = round(mean(Fare),2),
+         Passengers = n()
+     )
# A tibble: 2 x 3
  Survived Average_Fare Passengers
     <int>        <dbl>      <int>
1        0         23.0        424
2        1         51.8        290
```

Passengers who survived generally paid higher average fares compared to passengers who did not survive.

Insight:

Ticket fare is closely related to passenger class, which influenced access to safety resources.

5.5 Survival Based on Port of Embarkation

```
> titanic_cleaned %>%
+     group_by(Embarked) %>%
+     summarise(
+         Total = n(),
+         Survived = sum(Survived),
+         Survival_Rate = round(mean(Survived)*100,2)
+     )
# A tibble: 4 x 4
  Embarked Total Survived Survival_Rate
  <chr>    <int>    <int>         <dbl>
1 ""           2        2           200
2 "C"        130       79          7900
3 "Q"         28        8           800
4 "S"        554      201         20100
```

Passengers who embarked from Cherbourg had a higher survival rate compared to other embarkation ports.

Example observation:

Approximately 80% of passengers who embarked from Cherbourg survived, and around 25% of them were from Third Class.

This indicates that embarkation location may indirectly reflect passenger demographics or class distribution.

5.6 Family Size and Survival

```
> titanic_cleaned$FamilySize <- titanic_cleaned$SibSp + titanic_cleaned$Parch
>
> titanic_cleaned %>%
+     mutate(Travel_Type = ifelse(FamilySize == 0, "Alone", "With Family")) %>%
+     group_by(Travel_Type) %>%
+     summarise(
+         Total = n(),
+         Survived = sum(Survived),
+         Survival_Rate = round(mean(Survived)*100,2)
+     )
# A tibble: 2 x 4
  Travel_Type Total Survived Survival_Rate
  <chr>       <int>    <int>         <dbl>
1 Alone         404      130         13000
2 With Family   310      160         16000
```

Passengers travelling with family members had slightly higher survival chances compared to passengers travelling alone.

Insight:

Passengers travelling in groups may have had better coordination during evacuation.

6.0 CONCLUSION

This analysis demonstrates that survival on the Titanic was influenced by several factors, including gender, passenger class, ticket fare, and embarkation location.

Key findings include:

- Female passengers had significantly higher survival rates.
- First-class passengers were more likely to survive compared to third-class passengers.
- Higher ticket fares were associated with higher survival probability.
- Passengers who embarked from Cherbourg showed higher survival rates.

Overall, the analysis highlights how social class and demographic characteristics strongly influenced survival outcomes during the Titanic disaster.

7.0 TOOLS AND LIBRARIES USED

The following R libraries and functions were used in this analysis:

Library	**Purpose**
readxl	Import Excel files
dplyr	Data filtering and manipulation
base R functions	NA detection and dataset inspection

Key functions used include:

- read.csv()
- is.na()
- sapply()
- na.omit()
- filter()
- arrange()
- write.csv()